

SRI VASAVI ENGINEERING COLLEGE (Autonomous)**B.Tech V Semester Regular Examinations****Natural Language Processing**

(AIML)

Time: 3 Hrs

Max. Marks: 70

PART-A Answer All the Questions.				
1.				20 M
	a	Define Tokenization and explain its need in NLP.		CO1-K1(2M)
	b	What are Transducers? Give one application in NLP.		CO1-K1(2M)
	c	Write the main difference between Unsmoothed and Smoothed N-grams.		CO2-K1(2M)
	d	State any two limitations of Maximum Entropy Models.		CO2-K1(2M)
	e	What is a Treebank?		CO3-K1(2M)
	f	List two applications of Probabilistic CFGs.		CO3-K1(2M)
	h	Define Selectional Restrictions in semantics.		CO4-K1(2M)
	g	Write any two types of Relations between Senses.		CO4-K1(2M)
	h	List two advantages of using FrameNet.		CO5-K1(2M)
	i	Write two steps of Hobbs Algorithm for anaphora resolution.		CO5-K1(2M)
		PART-B All Questions Carry Equal Marks		
2.				10 M
	A.	i. Explain in detail English Morphology and its role in NLP.		CO1-K2
		ii. Describe the process of Tokenization with types and examples		CO1-K2
		OR		
	B.	i. Construct a Finite-State Automata that recognizes strings with even number of a's over {a, b}.		CO1-K2
		Write an algorithm for computing Minimum Edit Distance between two words.		CO1-K2
3.				10 M
	A.	i. Explain Smoothing techniques (Add-One, Good-Turing) with examples.		CO2-K2
		ii. Given the corpus 'please enjoy the meal', compute trigram probability P('enjoy the meal' 'please').		CO2-K3
		OR		
	B.	i. Discuss the architecture of a Hidden Markov Model.		CO2-K2
		ii. Prioritize Transformation-Based Tagging with an example.		CO2-K3
4.				10 M
	A.	i. Write the steps involved in Probabilistic CFG parsing		CO3-K2
		ii. Show Dynamic Programming Parsing using the CKY algorithm for a short grammar.		CO3-K3
		OR		
	B.	i. Differentiate between Context-Free Grammar and Dependency Grammar.		CO3-K2
		ii. Illustrate on Shallow Parsing in NLP.		CO3-K3
5.				10 M
	A.	i. Explain Syntax-Driven Semantic Analysis with an example.		CO4-K2
		ii. Apply Dictionary & Thesaurus method to disambiguate 'bass' in 'He likes to play bass'.		CO4-K3
		OR		
	B.	i. Illustrate First-Order Logic (FOL) representations for natural language sentences with suitable examples		CO4-K2
		ii. Compute Word Similarity between given vectors using Distributional Method.		CO4-K3
6.				10 M
	A.	i. Explain Coreference Resolution with an example.		CO5-K2
		ii. Summarize Porter Stemmer and its applications.		CO5-K3
		OR		
	B.	i. Describe the structure and uses of WordNet		CO5-K2

		ii.	Apply Centering Algorithm for Anaphora Resolution.	CO5-K3
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